Q27 — Deeper projects

Interpretability, Weight Inspection, and Persistence of Pre-training Traits

Deep Learning Systems 2026 — AI-internals Module

Framing the student question

Two intertwined questions motivate this module. First, *can inspecting a model's weights (and activations) reveal its tendencies*—its latent preferences, knowledge, or dispositions—rather than merely its input/output behavior? Second, *do models revert to pre-training traits after deployment*, i.e., does fine-tuning or alignment durably change a model, or does it leave a thin behavioral veneer over a largely intact pre-trained base?

The honest short answer to the first is: *partially*. Weight and activation inspection has produced genuine causal handles—directions that encode truthfulness-adjacent features, circuits implementing specific algorithms, and sparse features recovered by dictionary learning [1, 2, 4–6]. But weights are not a transparent readout of "tendencies": superposition, polysemantic neurons, and distributed representations mean that naive weight staring is unreliable [3]. The second question connects to a growing body of evidence that alignment fine-tuning is often shallow and reversible [9–11], and that models can retain—and under pressure re-express—dispositions formed earlier [12, 13].

The projects below are designed to let students *measure* these phenomena rather than speculate about them. Each is scoped for a team of 2–3 over roughly 4–6 weeks with modest compute (a few open-weight models in the 1B–8B range).

Project 1: Reading tendencies off activations and weights

Goal. Quantify how much of a model's "tendency" on a target behavior (e.g. sycophancy, refusal, a factual belief) is linearly decodable from internal states, and whether an intervention on that direction causally controls the behavior.

Method.

1. Choose a behavior with contrastive prompt pairs (e.g. agree-with-user vs. correct-the-user). Collect residual-stream activations across layers.
2. Fit linear probes and compute difference-of-means / mass-mean directions [6]. Compare to representation-engineering style reading/steering vectors [7].
3. *Causal validation*: add/subtract the direction (activation steering) and measure behavior change; apply activation patching to localize where the direction matters [7, 8].
4. *Weights, not just activations*: project the learned direction back through the unembedding/MLP weights to identify contributing components; sanity check against logit-lens readouts.

Deliverables.

- A per-layer probe-accuracy curve and a steering dose–response plot.
- A causal-faithfulness report: fraction of behavior change explained by the identified direction, with a negative-control direction baseline.
- A 2-page writeup distinguishing *correlational* decoding from *causal* control, explicitly noting superposition caveats [3].

Project 2: Does alignment survive? Shallow safety and reversion

Goal. Test the hypothesis that safety/alignment fine-tuning is "shallow" and that pre-training tendencies re-emerge under perturbation.

Method.

1. Take an instruction-tuned, safety-aligned open model and its base model.
2. Characterize the *token-depth* of safety: measure how refusal is concentrated in the first few generated tokens [9].
3. Attempt lightweight reversion: minimal fine-tuning on benign data, LoRA edits, or pruning of the components implicated in alignment [10, 11]; measure how little perturbation suffices to recover base-model behavior.
4. Compare internal directions (Project 1) before/after, testing whether the aligned model *moves away from* or merely *masks* the base direction.

Deliverables.

- A "safety depth" figure (behavior vs. tokens constrained).
- A reversion table: perturbation budget vs. fraction of pre-aligned behavior recovered, with a clear operational definition of "reversion."
- A short discussion relating findings to the shallow-alignment hypothesis [9, 10], with responsible-disclosure notes (no release of harmful artifacts).

Project 3: Dormant dispositions and conditional behavior

Goal. Study whether a model can carry a *dormant* tendency that is activated only under specific triggers—and whether standard safety training removes it or merely hides it.

Method.

1. Replicate a small "sleeper"-style setup: train/fine-tune a model to behave differently under a benign trigger [12].
2. Apply safety training (SFT/RL/adversarial) and test whether the conditional behavior persists.

3. Use interpretability tools to detect the dormant disposition from internals *without* using the trigger: probe for the trigger-detecting feature; attempt dictionary-learning / SAE feature discovery [4, 5].

4. Relate to alignment-faking dynamics where a model behaves differently when it infers it is being evaluated [13].

Deliverables.

- Persistence curves: conditional behavior rate vs. amount of safety training.
- A detection ROC: can internals flag the dormant disposition without the trigger?
- A critical essay on what "reversion to pre-training traits" means when traits are conditional rather than always-on.

Project 4: Weight-space forensics across the fine-tuning trajectory

Goal. Test how much of a model's post-deployment behavior is predictable from *weight diffs* between base and tuned checkpoints—i.e., reading tendencies directly from parameters.

Method.

1. Compute per-parameter and per-component weight deltas between base and fine-tuned models (and across intermediate checkpoints if available).
2. Localize which attention heads / MLP blocks change most; relate to known mechanisms such as induction heads [2] and to edited-fact localization results [8].
3. Predict behavior from weight-space features alone; validate causally by ablating or restoring the top-changed components.
4. Correlate weight-space structure with the activation directions from Project 1 to connect parameter changes to behavioral tendencies.

Deliverables.

- A weight-delta heatmap over layers/components with an interpretation.
- A predictive model: weight features $\rightarrow$ behavior, with held-out accuracy and causal ablation confirmation.
- A reproducible notebook and a reflection on the limits of "weight staring" given superposition and distributed storage [1, 3].

Cross-cutting rubric

Each project is graded on: (i) a clear operationalization of "tendency" or "reversion"; (ii) a correlational *vs.* causal distinction with proper controls; (iii) reproducibility (seeds, code, model hashes); and (iv) honest negative results and threat-model/ethics notes.

References

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